Engineering Mathematics - I · Unit_04 · Unit 4

DRAFT — UE26MA141A · Unit_04 · Unit 4 · Relation between Beta and Gamma functions, Duplication formula

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Relation between Beta and Gamma functions, Duplication formula

1 Relation between Beta and Gamma functions, Duplication formula

Study focus: Relation between Beta and Gamma functions, Duplication formula. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000030 · EV0000031

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In last class you have learnt definition and properties of beta function.

Key details

- In this class, you are going to learn relationship between beta gamma function and duplication formula.
- It is a very important relation which is very useful and it states that beta of m comma n is equal to gamma of n into gamma of n divided by gamma of m plus n.
- integral 0 to infinity e power minus y square y power to n minus 1 dy.
- So here in gamma m I have used x and in gamma n I have used y doesn't matter because we are dealing with definite integrals.
- That is equal to 4 times integral 0 to infinity 0 to infinity e power minus x square into e power minus y square which is e power minus of x square plus y square into x power two m minus one into y power two n minus one dx dy.
- Therefore, your x square plus y square is now r square and dx dy is r into dr into d theta.
- In Cartesian form, we know that x varies from 0 to infinity and y also varies from 0 to infinity.
- That means x and y both are positive.
- So, region of integration is the first quadrant.
- R cos theta whole power two m minus one into r sin theta whole power two n minus one into r into dr into d theta.

- So first let me write all the terms containing r together and then write all the terms containing theta together.
- E power minus r square r power two m minus one r power two n minus one r into cos power two m minus one theta sin power two n minus one theta into d r d theta.
- Here you can observe, so e power minus r square into r power 2n minus 1 into r power 2n minus 1 into r is purely a function of r and cos power 2n minus 1 theta into sin power 2n minus 1 theta is purely a function of theta.
- Two times integral zero to infinity e power minus r square r power two m minus one r power two n minus one r into dr.
- You can observe here the first integral is nothing but equal to beta m comma n.

Example or application from the lecture

- First, let us see what is relationship between beta gamma functions.
- First, we will consider the definition of gamma function.
- Then consider gamma n.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 05:00–09:59

e to the power minus t because we are putting r square is equal to t, so it is e power minus t into t to the power m minus one by two plus n minus one by two d t.

Key details

- We know that gamma n is equal to integral 0 to infinity e power minus t power n minus 1 dt.
- The expression whatever we got for gamma m into gamma n, which is beta of m, n into gamma m plus n.
- Thus we can conclude that beta of m comma n is equal to gamma m into gamma n divided by gamma of m plus n.
- It states that square root of pi into gamma of 2m is equal to 2 to the power 2m minus 1 into gamma of m into gamma of m plus 1 by 2.
- which is equal to two times integral 0 to pi by 2 sine power 2 n minus 1 theta cos power 2n minus 1 theta d theta.
- That means in the place of n I am putting m.
- Two variables there is only one variable.
- 2 m minus 1 theta d theta.
- That means 2 d theta is equal to dt.
- When theta is equal to 0, t is also equal to 0.
- When theta is equal to.
- Pi by two, your t is equal to pi.
- into dt by 2.
- And also you can use the result here 0 to pi sine power 2 m minus 1 t dt is equal to 2 times integral 0 to pi by 2 sin power 2 m minus 1 t.

- Thus my beta of m comma m now becomes two divided by two to the power two m minus one into integral zero to pi by two sine power two m minus one t dt, which is equal to one by two to the power two m minus one into beta of m comma half.
- That is by using the formula integral 0 to pi by 2 sine power p theta d theta is 1 by 2 into beta of p plus 1 by 2,1 by 2.
- We know that beta of m comma n is equal to gamma m into gamma n divided by gamma of m plus n.
- Apply this formula to the left hand side which is beta of m comma m so we get left hand side as gamma m into gamma m divided by gamma of m plus m which is gamma of two m.

Example or application from the lecture

- Next let us go for duplication formula for gamma function.
- To prove this, we first consider beta of m, n.
- we consider beta of m comma m.

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EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 10:00–14:59

Apply the same formula to the right hand side also, where 1 by 2 to the power 2m minus 1 is a constant and you have beta of m, half, so it is gamma half into gamma m divided by gamma of m plus half.

Key details

- We get gamma m divided by gamma 2m is equal to 1 by 2 to the power 2m minus 1 into square root of pi because we know gamma half is equal to square root of pi divided by gamma of m plus half.
- Thus we have square root of pi into gamma of 2m is equal to 2 to the power 2m minus 1 into gamma of m into gamma of m plus half.
- This is the required result.
- Since we need gamma n into gamma of 1 minus n,.
- And we know gamma 1 is equal to 1, so we have gamma n into gamma of 1 minus n is equal to beta of n, 1 minus n.
- And also we know beta of m comma n is equal to integral 0 to infinity x power n minus 1 divided by 1 plus x power m plus n dx.
- This is another form that is of a beta function.
- You have studied it.
- while studying the properties of beta function.
- Beta function as an improper integral.
- Now therefore we can have beta of n comma 1 minus n as.
- Integral zero to infinity x power n minus one divided by one plus x dx which is equal to pi divided by sin n pi.
- Why this is because of the fact that integral 0 to infinity x power n minus 1 divided by 1 plus x dx is equal to pi divided by sin n pi.

- The proof of this you will learn while studying the complex analysis.
- Thus, in this class you have studied relationship between beta gamma functions, duplication formula,.
- And also this result, which says gamma n into gamma of 1 minus n is equal to pi divided by sine n.
- With this we end this session.

Example or application from the lecture

- Next, we consider one important result that is gamma n into gamma of 1 minus n is equal to pi divided by sin n pi, where n lies between 0 and 1.
- To prove this, we consider the relationship between beta gamma function.
- Let us put M is equal to one minus N.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Relation between Beta and Gamma functions, Duplication formula transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/003_Relation_between_Beta_and_Gamma_functions_Duplication_formula__982d3dde7e270fc7.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Relation between Beta and Gamma functions, Duplication formula transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/003_Relation_between_Beta_and_Gamma_functions_Duplication_formula__982d3dde7e270fc7.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Relation between Beta and Gamma functions, Duplication formula transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/003_Relation_between_Beta_and_Gamma_functions_Duplication_formula__982d3dde7e270fc7.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9980; only the JSON segmentation supplies the displayed timestamps.
Relation between Beta and Gamma functions, Duplication formula transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/003_Relation_between_Beta_and_Gamma_functions_Duplication_formula__982d3dde7e270fc7.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00010	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0055. Sidecar variants compare...
SRC00011	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0055. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0055. Sidecar variants compare...
SRC00013	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0055. Sidecar variants compare...

C Revision and retrieval practice